



Determinants of University-Industry Knowledge Transfer Success in Iran: An Empirical PLS-SEM Model

Hamed Fazlollahtabar*

Department of Industrial Engineering, School of Engineering, Damghan University, 36716-45667 Damghan, Iran

* Correspondence: Hamed Fazlollahtabar (h.fazl@du.ac.ir)

Received: 10-09-2025

Revised: 11-26-2025

Accepted: 12-07-2025

Citation: H. Fazlollahtabar, "Determinants of university-industry Knowledge Transfer Success in Iran: An empirical PLS-SEM model," *Int J. Knowl. Innov Stud.*, vol. 3, no. 4, pp. 224–236, 2025. <https://doi.org/10.56578/ijkis030403>.



© 2025 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: University-Industry knowledge transfer plays a central role in converting academic research into technological and organizational innovation, but its success remains uneven in developing economies. This study investigates the factors associated with successful university-to-industry knowledge transfer in Iran and develops an empirical model for assessing such initiatives. Six explanatory constructs—Knowledge Destination, Knowledge Source, Knowledge Characteristics, University–Industry Distance, Transfer Mechanisms, and Government Dimension—were identified through a literature review and expert interviews. Survey data were then collected from 88 experts with experience in both academic and industrial settings and analyzed using partial least squares structural equation modeling (PLS-SEM). The results showed that Knowledge Characteristics, University–Industry Distance, Transfer Mechanisms, and Knowledge Destination had statistically significant positive relationships with Knowledge Transfer Success. In contrast, Knowledge Source and Government Dimension did not show statistically significant relationships with the outcome construct. The model explained 67% of the variance in Knowledge Transfer Success and demonstrated predictive relevance ($Q^2 = 0.36$). These findings indicate that successful knowledge transfer depends primarily on the nature and applicability of the knowledge, the distance between the participating organizations, the mechanisms used for transfer, and the receiving industry's capacity and readiness. The proposed framework provides an empirically tested basis for diagnosing weaknesses in University–Industry knowledge transfer projects and supports more focused decisions by university administrators, industry managers, and innovation policymakers.

Keywords: University-Industry knowledge transfer; Knowledge Transfer Success; Transfer Mechanisms; Organizational distance; Innovation policy; Partial least squares structural equation modeling

1 Introduction

University-Industry relationships provide an important channel through which academic knowledge enters industrial practice. In developed economies, sustained interaction between universities and firms has supported technological development, industrial renewal, and economic growth. In many developing economies, however, these relationships remain fragmented or weak. Research findings generated within universities may consequently have limited industrial application, even when they address practical production, managerial, or technological problems. Weak University-Industry relationships are therefore one source of unsuccessful knowledge transfer, although the success or failure of the process also depends on the characteristics of the knowledge, the participating organizations, the transfer arrangements, and the surrounding institutional environment [1].

Knowledge transfer refers to the movement of knowledge from a source to a recipient. It includes the sharing of tacit knowledge as well as the communication, interpretation, and application of explicit knowledge between individuals, groups, or organizations. Cooperation between universities and industries is particularly important in a knowledge-based economy because universities produce research findings, technical expertise, and new ideas, while industries provide settings in which this knowledge can be tested, adapted, and applied. The demand for such cooperation has grown as firms increasingly look beyond their internal research and development activities for knowledge that can support products, processes, and organizational practices. At the same time, universities and research institutions face growing expectations to demonstrate that their research has value outside academic settings. These changes have altered the conventional division of roles between knowledge producers and knowledge

users. Successful transfer therefore requires more than the availability of research results; it also requires suitable organizational conditions, receptive industrial partners, effective Transfer Mechanisms, and continued interaction between the source and the recipient. Identifying these conditions is central to understanding why some knowledge transfer projects produce usable outcomes while others fail to move beyond the initial exchange of information [2, 3].

For firms operating in rapidly changing markets, access to external knowledge is not sufficient by itself. They must also be able to recognize its relevance, absorb it, adapt it to their operational needs, and incorporate it into existing processes. Universities, in turn, must be willing and able to present research outputs in forms that industrial partners can understand and use. Difficulties may arise when the knowledge is overly complex, insufficiently documented, weakly connected to industrial needs, or communicated through unsuitable channels. Differences in organizational objectives, professional language, institutional culture, geographical location, and expectations may further obstruct the process. Some recipient organizations may also lack the internal capacity, resources, or managerial commitment needed to support knowledge absorption. Universities and industries must therefore identify the conditions that strengthen or weaken transfer and allocate their responsibilities accordingly. Such an assessment can reduce avoidable failures and provide a clearer basis for managing knowledge transfer projects [4].

This issue is especially relevant in Iran, where universities produce a substantial body of scientific and technical research, but the direct application of academic knowledge in industry remains uneven. Examining this setting requires attention to both sides of the transfer relationship and to the knowledge being transferred. It also requires consideration of the distance between universities and industries, the mechanisms through which knowledge is communicated, and the role of government. Treating University-Industry knowledge transfer as a multidimensional process makes it possible to examine these conditions within a single empirical model rather than considering them separately.

Against this background, the present study investigates the determinants of university-to-industry Knowledge Transfer Success in Iran. It develops and tests a conceptual model comprising six explanatory constructs: Knowledge Destination, representing the characteristics and readiness of the receiving industry; Knowledge Source, representing the capabilities and conduct of the university; Knowledge Characteristics; University-Industry Distance; Transfer Mechanisms; and Government Dimension. Knowledge Transfer Success is specified as the outcome construct. The study has three objectives:

- To identify and organize the factors associated with knowledge transfer from universities to industries and use them to construct a conceptual framework.
- To estimate the relationships between the identified factors and Knowledge Transfer Success and compare their relative importance.
- To provide evidence that can inform policymakers, university administrators, industry managers, and other stakeholders involved in University-Industry knowledge transfer.

Accordingly, the study addresses the following research questions:

1. Which source-related, recipient-related, knowledge-related, relational, transfer-related, and governmental factors are associated with successful university-to-industry knowledge transfer?
2. Do these factors differ in the direction, magnitude, and statistical significance of their relationships with Knowledge Transfer Success?
3. Does the proposed model demonstrate adequate explanatory and predictive performance when applied to expert evidence from the Iranian University-Industry context?

Existing studies have examined knowledge transfer from several perspectives, but their coverage remains uneven. A substantial part of the literature concentrates on knowledge exchange within organizations, including transfers among individuals, groups, organizational units, and research and development departments. Other studies examine exchanges between R&D centers or address the determinants of technology transfer. Although technology transfer includes a knowledge component, it does not fully represent the movement of scientific findings, ideas, expertise, and research results from universities to industrial users. Some studies have discussed the advantages associated with knowledge transfer without empirically examining the conditions responsible for successful transfer [5]. Others have considered the value of University-Industry relationships or described the forms that such interactions may take, but have not integrated the characteristics of the source, destination, knowledge, organizational distance, transfer mechanism, and government environment into a single testable framework.

The present study addresses this gap by developing a factor-oriented model specifically for university-to-industry knowledge transfer and testing it through partial least squares structural equation modeling (PLS-SEM). The model brings together organizational, relational, knowledge-related, procedural, and institutional dimensions and estimates their respective relationships with transfer success. In doing so, the study distinguishes knowledge transfer from the broader concept of technology transfer and directs attention to the conditions under which academic knowledge becomes understandable, accessible, and usable in an industrial setting. Its contribution to knowledge and innovation studies lies in connecting the production of academic knowledge with its organizational reception and practical use. The findings also provide a structured basis for identifying weaknesses in knowledge transfer projects and for directing

managerial and policy attention toward the factors supported by the empirical results.

2 Literature Review

Information and communication technologies have substantially changed how research findings are stored, disseminated, and accessed. Organizations can now obtain academic publications and technical information more readily than in the past. Easier access, however, does not necessarily lead to the practical use of academic knowledge. A persistent gap remains between the knowledge produced by universities and the knowledge that organizations can understand, absorb, and apply. Research [6] attributes part of this gap to the greater attention researchers often give to producing knowledge than to communicating their findings to non-academic users. When research results are not translated into accessible and operational forms, managers, policymakers, and other stakeholders may find it difficult to recognize their relevance or apply them in practice.

Barriers may also arise on the receiving side. Organizations sometimes resist unfamiliar knowledge or established practices that require operational change [7]. They may also lack the personnel, technical skills, organizational routines, or absorptive capacity needed to interpret and use research findings [8]. The gap between knowledge creation and application therefore cannot be attributed exclusively to either universities or industries. It reflects conditions on both sides of the relationship, together with the nature of the knowledge and the arrangements through which it is transferred.

A firm's technical knowledge base supports the development of new products, production processes, and internal practices. Such knowledge is not acquired solely through internal learning. Firms increasingly draw on universities and other external sources to obtain scientific and technical knowledge that is unavailable within their existing organizational boundaries. Establishing a workable relationship with an external knowledge source can consequently shape a firm's ability to compete in environments characterized by rapid technological change [9]. Universities also have reasons to participate in these relationships. Alongside teaching and academic publication, the production and external use of relevant research may affect their public role, institutional standing, and contribution to economic and social development [10]. University-Industry knowledge transfer must therefore be understood as a reciprocal process in which the source and recipient identify, communicate, interpret, and apply knowledge under specific organizational and institutional conditions.

2.1 Levels and Models of Knowledge Transfer

Knowledge transfer may take place at four levels [11]:

Individual level: At the most basic level, knowledge is received and interpreted by an individual, reconstructed in relation to that person's experience, and subsequently communicated to another individual. Even when transfer occurs between organizations, individuals remain directly involved in interpreting and applying the knowledge.

Intra-organizational level: Knowledge is transferred between units within the same organization. One example is the movement of knowledge from an R&D unit, where new technical knowledge is produced, to a production unit, where it is absorbed and applied.

Inter-organizational level: Knowledge moves from one organization to another. University-to-industry knowledge transfer belongs primarily to this level because an industrial recipient obtains knowledge from an external academic source within the national environment.

Transnational level: Knowledge is transferred across national boundaries. Such projects may involve additional geographical, institutional, cultural, legal, and linguistic differences between the source and recipient.

This classification identifies where knowledge transfer occurs, but it does not by itself explain how the process operates or why it succeeds. The literature has therefore developed several types of knowledge transfer models. Based on the reviewed literature, these models can be divided into eight categories [12]:

1. Process-oriented models.
2. Factor-oriented models.
3. Factor and process-oriented models.
4. Element-oriented models.
5. Individual knowledge transfer models.
6. Intra-organizational knowledge transfer models.
7. Inter-organizational knowledge transfer models.
8. International knowledge transfer models.

Process-oriented models describe the stages through which knowledge moves from its source to its eventual application. Factor-oriented models identify the conditions associated with successful or unsuccessful transfer. Models combining factors and processes examine both the stages of transfer and the conditions affecting those stages. Element-oriented models focus on the principal components involved in an exchange. The remaining four categories distinguish models according to the level at which knowledge transfer occurs.

The present study concerns inter-organizational knowledge transfer and adopts a factor-oriented approach. This approach is appropriate because the purpose is to identify the conditions associated with Knowledge Transfer Success and estimate their respective relationships with that outcome.

2.2 Elements of Knowledge Transfer

Across the different model categories, the knowledge transfer process can be described through four basic elements:

Knowledge actors: These are the individuals and organizations participating in the process, including the knowledge source, the recipient, and any intermediary that assists communication or application.

Context: This refers to the organizational, relational, geographical, cultural, and institutional conditions under which the source and recipient interact.

Content: This is the knowledge being transferred. Its complexity, applicability, relevance, form, and degree of documentation may affect whether the recipient can understand and use it.

Tools: These are the channels, procedures, activities, and mechanisms through which knowledge is communicated and transferred.

These elements are closely connected. The characteristics of the actors affect their capacity and willingness to participate, while contextual differences may facilitate or obstruct their relationship. At the same time, the content of the knowledge determines how difficult it is to communicate, and the selected tools determine whether the transfer process is suitable for the actors and the knowledge concerned.

Following a review of studies on University-Industry and inter-organizational knowledge transfer, the factors identified in the literature were organized according to these elements and their related dimensions. Table 1 presents the extracted factors, their dimensions, and their supporting sources.

Table 1. Extracted factors based on literature review

Factor	Dimension	Source
Knowledge complexity	Knowledge characteristics	[13–15]
Tacit knowledge	Knowledge characteristics	[13, 14]
Degree of knowledge applicability	Knowledge characteristics	[16, 17]
Availability of basic knowledge	Knowledge characteristics	[17, 18]
Degree of knowledge relevance	Knowledge characteristics	[17, 18]
Explicit and documented knowledge	Knowledge characteristics	[17]
Organizational distance	Relationship dimension	[13, 14]
Knowledge distance between source and receiver	Relationship dimension	[13, 14, 19]
Physical distance	Relationship dimension	[13, 14, 20]
Normative distance	Relationship dimension	[13, 14]
Mutual trust	Relationship dimension	[14, 19]
Source-receiver relationships	Relationship dimension	[19, 21]
Mutual distance	Relationship dimension	[22]
Cultural awareness	Relationship dimension	[19]
Goals and focus	Relationship dimension	[19]
Knowledge transfer speed	Transfer mechanism and relationship dimension	[13, 14]
Transfer activities	Transfer mechanism	[13, 14]
Transfer Mechanisms and tools	Transfer mechanism	[17, 19, 23]
Clear and precise language	Transfer mechanism	[19, 23]
Receiver's learning culture	Receiver dimension	[13, 14]
Project priority	Receiver dimension	[17]
Receiver's credibility	Receiver dimension	[14]
Absorption capacity	Receiver dimension	[19]
Receiver's willingness to transfer	Receiver dimension	[17]
Receiver's ability to transfer	Receiver dimension	[17]
Source's willingness to transfer	Source dimension	[17]
Source's ability to transfer	Source dimension	[17]
Source's credibility	Source dimension	[17]
Source's motivation to teach the receiver	Source dimension	[17, 19, 23]
Knowledge translation	Source and receiver dimension	[23]

2.3 Factors Associated With University-Industry Knowledge Transfer

The transfer of academic knowledge to industry can contribute to changes in products, methods, procedures, processes, and organizational policies. However, the production of relevant knowledge does not guarantee its

industrial use. The reviewed literature generally organizes the conditions affecting knowledge transfer into five categories: (1) knowledge characteristics, (2) source characteristics, (3) the relationship between the source and recipient, (4) transfer activities and mechanisms, and (5) recipient characteristics [13, 14].

The knowledge characteristics dimension concerns the form and properties of the content being transferred. Academic knowledge may differ in complexity, applicability, relevance, documentation, and dependence on prior knowledge. Complex or largely tacit knowledge generally requires closer interaction and more extensive interpretation than explicit and documented knowledge. When recipients do not possess the necessary background knowledge, they may be unable to understand or adapt technically sound research. Applicability and relevance are equally important because knowledge that does not correspond to an identifiable industrial problem is unlikely to be adopted.

The source dimension refers to the university's ability, willingness, credibility, and motivation to communicate its knowledge. A university may produce valuable research but still encounter transfer difficulties if researchers have little interest in industrial engagement, lack suitable communication skills, or cannot translate scientific findings into forms that industrial users can interpret.

The relationship dimension covers the distance between the university and industry. Distance is not limited to geographical separation. It may also involve organizational practices, professional norms, knowledge bases, institutional expectations, goals, and culture. Mutual trust and continued source–recipient interaction can reduce some of these differences, whereas weak relationships may make it difficult to exchange tacit knowledge or resolve misunderstandings.

The transfer mechanism dimension concerns the activities, channels, tools, language, and speed of knowledge transfer. The selected mechanism must correspond to the nature of the knowledge and the capabilities of the recipient. Written documents may be sufficient for explicit knowledge, while tacit or technically complex knowledge may require meetings, demonstrations, training, joint work, or continued consultation.

The recipient dimension concerns the industrial organization's learning culture, absorptive capacity, project priorities, credibility, willingness, and ability to acquire and apply external knowledge. A recipient may have access to relevant academic findings but fail to use them if the project lacks managerial support, qualified personnel, financial resources, or operational priority.

In the empirical model used in this study, these categories were reorganized into Knowledge Characteristics, Knowledge Source, University-Industry Distance, Transfer Mechanisms, and Knowledge Destination. The recipient-related factors identified in the literature form the Knowledge Destination construct, while relational factors are represented primarily by University-Industry Distance and, where applicable, the relational components of the transfer mechanism. Table 1 provides the detailed basis for this classification.

The factors in Table 1 show that University-Industry knowledge transfer is not a single act of communication. It is a coordinated process shaped by the content being transferred, the capabilities of the university and industrial recipient, the distance between them, and the mechanisms used to connect the two organizations. These dimensions provide the theoretical basis for the constructs evaluated in the partial least squares structural equation modeling (PLS-SEM) model.

3 Research Methodology

3.1 Research Design

This study adopted a quantitative research design to develop and empirically examine a model of university-to-industry Knowledge Transfer Success in Iran. The proposed model was constructed through a review of the relevant literature and consultations with experts familiar with knowledge transfer in academic and industrial settings. The literature review identified the principal factors associated with inter-organizational knowledge transfer, while the expert consultations were used to examine their relevance to University-Industry relationships in the Iranian context.

The resulting model comprised six explanatory constructs: Knowledge Destination, Knowledge Source, Knowledge Characteristics, University-Industry Distance, Relational Dimension and Transfer Mechanism, and Government Dimension. Knowledge Transfer Success was specified as the endogenous construct. The model was designed to determine which of the six constructs were significantly associated with transfer success and to evaluate its explanatory and predictive performance.

3.2 Instrument Development and Data Collection

A structured questionnaire was developed from the factors identified through the literature review and expert consultations. The initial instrument contained 34 indicators distributed across the seven constructs in the conceptual model. Knowledge Destination represented the characteristics and readiness of the industrial recipient, while Knowledge Source reflected the characteristics of the university providing the knowledge. Knowledge Characteristics covered the complexity, applicability, relevance, and form of the knowledge being transferred. University-Industry Distance represented the organizational, knowledge-related, physical, normative, and cultural differences between the two parties. Relational Dimension and Transfer Mechanism concerned the relationships, activities, channels,

and tools through which knowledge was communicated. Government Dimension represented the institutional and governmental conditions surrounding the transfer process. Knowledge Transfer Success captured the perceived outcomes of the transfer project.

Questionnaire data were collected from 88 experts who had experience in both academic and industrial environments. The use of respondents familiar with both sides of the transfer relationship allowed the proposed factors to be evaluated from an inter-organizational perspective. The completed questionnaires were screened and prepared in Excel before the measurement and structural models were examined using the PLS-SEM.

3.3 Data Analysis

PLS-SEM was selected because the study sought to examine relationships among several latent constructs while assessing both the measurement properties of the questionnaire and the explanatory performance of the structural model. The analysis proceeded in two stages. First, the measurement model was assessed through indicator loadings, internal consistency reliability, and convergent validity. Indicators with inadequate loadings were removed before the model was re-estimated. Second, the structural model was assessed using the estimated effects, bootstrap *t*-values, the coefficient of determination (R^2), and Stone–Geisser’s predictive relevance criterion (Q^2).

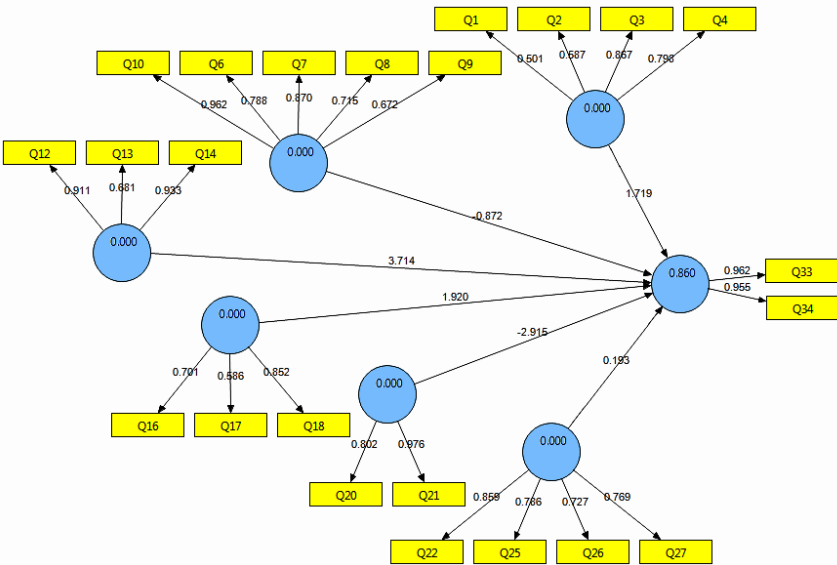


Figure 1. Structural model showing the indicator loadings for the measurement constructs

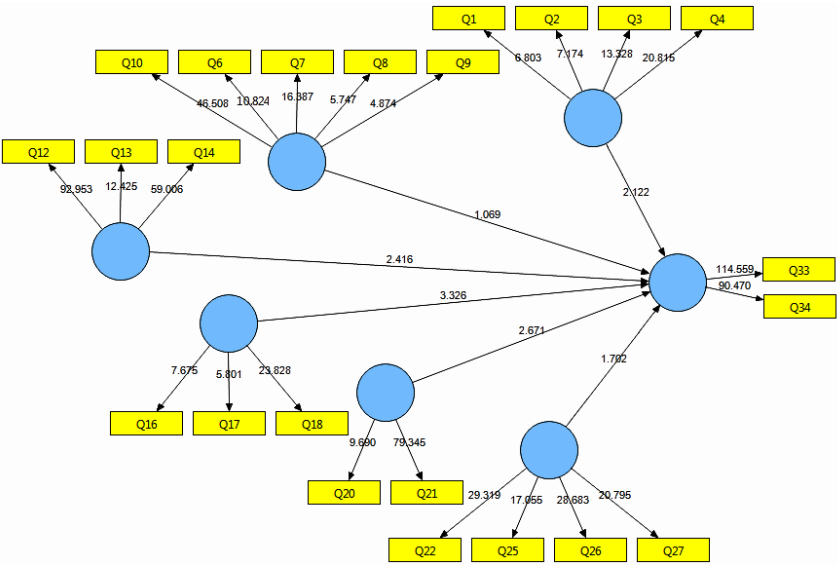


Figure 2. Structural model showing the estimated effects and corresponding *t*-values

Figure 1 and Figure 2 present the estimated PLS-SEM model. Figure 1 shows the measurement relationships between the indicators and their corresponding constructs. Figure 2 presents the structural relationships between the six explanatory constructs and Knowledge Transfer Success, together with their associated *t*-values.

4 Results

4.1 Initial Indicator Assessment

The initial model contained 34 indicators. Indicator loadings were examined to determine how well each item represented its assigned construct. As reported in Table 2, most indicators had satisfactory loadings, but several items produced low or anomalous values. These included Q5, Q11, Q19, Q23, Q24, Q28, Q29, Q30, Q31, and Q32. Q15 produced a negative loading and was also excluded from the re-estimated model. The indicators that did not meet the adopted measurement requirements were removed before the PLS-SEM analysis was repeated.

Table 2. Initial indicator loadings before item screening

Indicator	Factor	Factor Loading
Q1	Knowledge Destination	0.550
Q2	Knowledge Destination	0.559
Q3	Knowledge Destination	0.848
Q4	Knowledge Destination	0.776
Q5	Knowledge Destination	0.316
Q6	Knowledge Source	0.792
Q7	Knowledge Source	0.870
Q8	Knowledge Source	0.713
Q9	Knowledge Source	0.672
Q10	Knowledge Source	0.962
Q11	Knowledge Characteristics	0.350
Q12	Knowledge Characteristics	0.898
Q13	Knowledge Characteristics	0.727
Q14	Knowledge Characteristics	0.919
Q15	Distance Between University and Industry	-0.702
Q16	Distance Between University and Industry	0.733
Q17	Distance Between University and Industry	0.541
Q18	Distance Between University and Industry	0.875
Q19	Relational Dimension and Transfer Mechanism	0.264
Q20	Relational Dimension and Transfer Mechanism	0.743
Q21	Relational Dimension and Transfer Mechanism	0.917
Q22	Relational Dimension and Transfer Mechanism	0.697
Q23	Government Dimension	-0.088
Q24	Government Dimension	-0.027
Q25	Government Dimension	0.707
Q26	Government Dimension	0.808
Q27	Government Dimension	0.699
Q28	Government Dimension	-0.001
Q29	Government Dimension	-0.445
Q30	Government Dimension	0.285
Q31	Knowledge Transfer Success	0.000
Q32	Knowledge Transfer Success	0.370
Q33	Knowledge Transfer Success	0.939
Q34	Knowledge Transfer Success	0.971

4.2 Revised Measurement Model

The model was re-estimated after the indicators that did not meet the adopted measurement requirements had been excluded. Table 3 presents the loadings and bootstrap *t*-values for the indicators retained in the revised model. The retained loadings ranged from 0.501 to 0.976, and all corresponding *t*-values exceeded 1.96. These results indicated that the retained indicators were statistically associated with their respective constructs.

The lowest retained loading was recorded for Q1 under Knowledge Destination (0.501), followed by Q17 under University-Industry Distance (0.586) and Q2 under Knowledge Destination (0.587). Although these values were lower than those of the remaining indicators, they exceeded the minimum criterion adopted in the analysis. The other retained indicators showed moderate to strong loadings.

Table 3. Indicator loadings and bootstrap *t*-values for the revised measurement model

Construct	Indicator	Factor Loading	<i>t</i> -Value
Knowledge Destination	Q1	0.501	7.457
Knowledge Destination	Q2	0.587	6.556
Knowledge Destination	Q3	0.867	16.520
Knowledge Destination	Q4	0.798	11.916
Knowledge Destination	Q6	0.788	9.942
Knowledge Source	Q7	0.870	16.857
Knowledge Source	Q8	0.715	6.440
Knowledge Source	Q9	0.672	5.880
Knowledge Source	Q10	0.962	43.288
Knowledge Characteristics	Q12	0.911	106.650
Knowledge Characteristics	Q13	0.681	13.807
Knowledge Characteristics	Q14	0.933	75.180
Distance Between University and Industry	Q16	0.701	7.186
Distance Between University and Industry	Q17	0.586	4.227
Distance Between University and Industry	Q18	0.852	22.076
Relational Dimension and Transfer Mechanism	Q20	0.802	12.993
Relational Dimension and Transfer Mechanism	Q21	0.976	77.743
Relational Dimension and Transfer Mechanism	Q22	0.859	29.974
Government Dimension	Q25	0.786	18.841
Government Dimension	Q26	0.727	23.976
Government Dimension	Q27	0.769	23.149
Knowledge Transfer Success	Q33	0.962	105.404
Knowledge Transfer Success	Q34	0.955	87.200

4.3 Internal Consistency and Convergent Validity

The measurement model was assessed using Cronbach's alpha, composite reliability, and average variance extracted. As shown in Table 4, composite reliability ranged from 0.823 to 0.962 and exceeded 0.70 for every construct. The AVE values ranged from 0.602 to 0.928 and were therefore above the commonly applied threshold of 0.50. These results indicated that the retained indicators accounted for an acceptable proportion of variance in their respective constructs.

Cronbach's alpha exceeded 0.70 for six of the seven constructs. University-Industry Distance had a Cronbach's alpha of 0.684, which was slightly below 0.70. However, its composite reliability was 0.823 and its AVE was 0.609. The internal consistency of this construct was therefore regarded as marginal but acceptable in the context of the present exploratory model. Taken together, the reported statistics supported the internal consistency and convergent validity of the revised measurement model.

Table 4. Internal consistency reliability and convergent validity

Construct	Cronbach's Alpha	CR	AVE
Knowledge Destination	0.821	0.883	0.602
Knowledge Source	0.862	0.907	0.709
Knowledge Characteristics	0.845	0.907	0.766
Distance Between University and Industry	0.684	0.823	0.609
Relational Dimension and Transfer Mechanism	0.844	0.906	0.762
Government Dimension	0.835	0.891	0.672
Knowledge Transfer Success	0.952	0.962	0.928

Note: CR = Composite Reliability; AVE = Average Variance Extracted.

4.4 Structural Model Results

After the measurement model had been assessed, the relationships between the six explanatory constructs and Knowledge Transfer Success were examined. Because the model contained no mediating variables, the reported total-effect estimates represented the direct relationships between the explanatory constructs and Knowledge Transfer Success.

As shown in Table 5, Knowledge Destination had a statistically significant positive relationship with Knowledge Transfer Success (estimate = 1.72, *t* = 1.971). Knowledge Characteristics was also positively and significantly related

to Knowledge Transfer Success (estimate = 3.71, $t = 2.239$). Significant positive relationships were further observed for University-Industry Distance (estimate = 1.92, $t = 3.279$) and Relational Dimension and Transfer Mechanism (estimate = 2.92, $t = 2.787$).

Knowledge Source did not show a statistically significant relationship with Knowledge Transfer Success (estimate = -0.87, $t = 1.131$). Government Dimension was also statistically nonsignificant at the 5% level (estimate = 0.192, $t = 1.716$). The absence of statistical significance does not establish that these two constructs have no practical role. It indicates that their relationships with Knowledge Transfer Success were not supported by the present sample and model specification.

Several reported estimates exceeded an absolute value of 1.00. They are therefore presented as total-effect estimates rather than used to establish a definitive ranking of the constructs. Their magnitudes should be interpreted cautiously and alongside the bootstrap results, particularly because correlations among the explanatory constructs may affect individual estimates.

Table 5. Total-effect estimates and significance tests

Dependent Variable	Independent Variable	Total Effects	t -Value	Significance
Knowledge Transfer Success	Knowledge Destination	1.72	1.971	Significant
	Knowledge Source	-0.87	1.131	Not significant
	Knowledge Characteristics	3.71	2.239	Significant
	Distance Between University and Industry	1.92	3.279	Significant
	Relational Dimension and Transfer Mechanism	2.92	2.787	Significant
	Government Dimension	0.192	1.716	Not significant

4.5 Explanatory and Predictive Performance

The coefficient of determination showed that the six explanatory constructs jointly accounted for 67% of the variance in Knowledge Transfer Success ($R^2 = 0.67$). This result indicated substantial explanatory performance within the analyzed sample. The Stone–Geisser value was positive ($Q^2 = 0.36$), indicating that the model had predictive relevance for the endogenous construct.

The reported redundancy value was 0.33. The original analysis also produced a global Goodness of Fit (GoF) value of 0.76. Because GoF statistics do not replace separate assessments of the measurement and structural models in PLS-SEM, this value was treated as supplementary information rather than as independent proof of model validity (Table 6).

Table 6. Supplementary model assessment criteria

Fit Criterion	Value	Status
Redundancy	0.33	Suitable
Overall model fit (GoF)	0.76	Strong

Note: GoF = Goodness of Fit.

The single f^2 value reported in Table 7 describes the result available from the original analysis but does not permit a factor-by-factor ranking. Accordingly, the six explanatory constructs were not ranked on the basis of this value. Construct-specific effect sizes would be required to compare their individual contributions.

Table 7. Explanatory and predictive criteria

Evaluation Criterion	Value	Reference Values
R^2 (Coefficient of determination)	0.67	Strong (0.67), moderate (0.33), weak (0.19)
f^2 (Effect size)	0.35	Strong (0.35), moderate (0.15), weak (0.10)
Q^2 (Stone-Geisser)	0.36	Strong (0.36), moderate (0.25), weak (0.10)
GoF (Goodness of Fit)	0.76	Strong (0.36), moderate (0.25), weak (0.10)

4.6 Answers to the Research Questions

The empirical analysis addressed the three research questions concerning the factors associated with university-to-industry knowledge transfer, differences among their relationships with transfer success, and the performance of the proposed model.

Research Question 1: Which factors influence the university-to-industry knowledge transfer process?

The conceptual model identified six groups of factors relevant to university-to-industry knowledge transfer in Iran: Knowledge Destination, Knowledge Source, Knowledge Characteristics, University-Industry Distance, Relational Dimension and Transfer Mechanism, and Government Dimension. These constructs represented the industrial recipient, the academic source, the knowledge being transferred, the distance between the participating organizations, the arrangements used to conduct the transfer, and the surrounding governmental environment.

The empirical results showed that Knowledge Destination, Knowledge Characteristics, University-Industry Distance, and Relational Dimension and Transfer Mechanism had statistically significant relationships with Knowledge Transfer Success. Knowledge Source and Government Dimension did not show statistically significant relationships in the present analysis.

Research Question 2: Do the identified factors have the same relationship with Knowledge Transfer Success?

The factors differed in both the direction and statistical significance of their estimated relationships with Knowledge Transfer Success. Four constructs had significant positive relationships with the outcome construct. Knowledge Source produced a negative but statistically nonsignificant estimate, while Government Dimension produced a positive but statistically nonsignificant estimate.

The results therefore did not support the conclusion that all six constructs had the same degree of empirical importance. However, a definitive ranking of the constructs was not made because construct-specific effect sizes were unavailable and several reported total-effect estimates exceeded 1.00.

Research Question 3: Does the proposed model demonstrate adequate explanatory and predictive performance?

The model explained 67% of the variance in Knowledge Transfer Success, and the positive Q^2 value indicated predictive relevance within the analyzed sample. The retained indicators also showed generally acceptable internal consistency and convergent validity. These findings support the use of the framework as an exploratory means of examining university-to-industry knowledge transfer in Iran.

The model converts expert assessments into structured information that can assist the examination of knowledge transfer projects. It does not replace managerial or policy judgment. Decisions concerning a particular project should also consider its industrial setting, institutional arrangements, available resources, and possible consequences. Further testing with larger samples, separate university and industry respondent groups, and evidence from other national settings is required before the model can be treated as broadly generalizable.

5 Conclusion and Recommendations

5.1 Conclusion

This study developed and empirically examined a PLS-SEM model of university-to-industry Knowledge Transfer Success in Iran. The model considered six explanatory constructs: Knowledge Destination, Knowledge Source, Knowledge Characteristics, University-Industry Distance, Relational Dimension and Transfer Mechanism, and Government Dimension. Knowledge Transfer Success was specified as the outcome construct. The analysis was based on questionnaire responses from 88 experts with experience in academic and industrial settings.

The results showed that Knowledge Destination, Knowledge Characteristics, University-Industry Distance, and Relational Dimension and Transfer Mechanism had statistically significant positive relationships with Knowledge Transfer Success. These findings indicate that successful transfer depends not only on the production of academic knowledge but also on its relevance and applicability, the readiness of the industrial recipient, the distance between the participating organizations, and the mechanisms through which the knowledge is communicated and applied.

Knowledge Source and Government Dimension did not show statistically significant relationships with Knowledge Transfer Success in the present model. These results should not be interpreted as evidence that universities and government institutions are unimportant. Instead, they suggest that the effects of these two dimensions were not established within the present sample and model specification. Their roles may be indirect, dependent on particular institutional conditions, or represented differently across industries and knowledge transfer projects.

The model explained 67% of the variance in Knowledge Transfer Success, while the positive Stone-Geisser value ($Q^2 = 0.36$) indicated predictive relevance within the analyzed sample. The results therefore provide empirical support for treating University-Industry knowledge transfer as a multidimensional process involving the knowledge itself, the source and recipient organizations, their relationship, the transfer arrangements, and the institutional setting.

The principal contribution of this study lies in integrating these dimensions into a single framework for examining university-to-industry knowledge transfer in Iran. Unlike approaches that treat knowledge transfer as a simple movement of information, the proposed framework considers whether academic knowledge can be understood, received, adapted, and applied by an industrial organization. It also distinguishes knowledge transfer from the

broader concept of technology transfer by focusing on scientific findings, expertise, ideas, and research outputs as the content of the exchange.

The framework can be used as an initial diagnostic structure for identifying possible weaknesses in knowledge transfer projects. However, it should support rather than replace managerial and policy judgment. Its application to a specific project requires consideration of the industry concerned, the type of knowledge being transferred, the capabilities of the participating organizations, and the wider institutional environment.

5.2 Research Limitations

Several limitations should be considered when interpreting the findings. First, the analysis was based on 88 respondents. Although the participants had experience in both university and industrial settings, the sample size limits the precision of the estimates and the extent to which the results can be generalized.

Second, the study was conducted in Iran. University-Industry relationships are shaped by national innovation policies, institutional structures, funding arrangements, labor markets, and industrial conditions. The relationships identified in this study may therefore differ in other countries or institutional environments.

Third, the respondents were professors from leading Iranian universities who had experience in university-to-industry knowledge transfer projects. Their familiarity with both settings was relevant to the research purpose, but the sample did not provide an independent and balanced comparison of university and industry perspectives. Industrial managers, technical personnel, government officials, and knowledge-transfer intermediaries may evaluate the same factors differently.

Fourth, the study relied on questionnaire responses collected at one point in time. The results represent expert assessments rather than longitudinal observations of completed knowledge transfer projects. Consequently, the model cannot establish causal relationships or determine how the importance of individual factors changes across different stages of a project.

Fifth, several indicators were removed during measurement-model assessment. Although the retained indicators showed generally acceptable reliability and convergent validity, removing items may have narrowed the conceptual coverage of some constructs. The University-Industry Distance construct also produced a Cronbach's alpha slightly below 0.70 and should be examined further.

Finally, several total-effect estimates were larger than 1.00. These values may reflect substantial relationships among the explanatory constructs, suppression effects, or model-specification issues. The relative magnitude of the constructs should therefore be interpreted cautiously until the model has been tested using larger and more diverse samples and accompanied by detailed collinearity and construct-specific effect-size assessments.

5.3 Directions for Future Research

Future studies should test the proposed model with larger samples drawn from a wider range of universities, industries, and geographical regions. Collecting separate samples from academic researchers and industry practitioners would make it possible to compare the two sides of the transfer relationship and determine whether they evaluate the determinants of success differently.

The model could also be examined separately across manufacturing and service industries. The relevance, complexity, and transferability of academic knowledge may vary considerably between sectors. Similar comparisons could be conducted across academic disciplines, particularly between engineering, natural sciences, medicine, management, and the social sciences.

Longitudinal research is needed to follow knowledge transfer projects from their initiation to the application of research results. Such designs could determine whether different factors matter at the stages of knowledge identification, communication, adaptation, implementation, and continued use. They could also connect expert assessments with observable project outcomes.

Further research should examine the possible indirect roles of Knowledge Source and Government Dimension. For example, government policy may influence transfer success through funding, intellectual property arrangements, intermediary organizations, or Transfer Mechanisms rather than through a direct relationship. The influence of the university may likewise depend on the characteristics of the knowledge, the quality of the relationship, or the recipient's absorptive capacity. Testing mediation and moderation relationships would provide a more detailed account of these mechanisms.

Researchers may also adapt the framework to knowledge transfer between other types of organizations, including research institutes, government agencies, private firms, nonprofit organizations, and international partners. In addition, the measurement scales should be reassessed to confirm the content and discriminant validity of the constructs and to determine whether the indicators removed in the present study should be revised or replaced.

5.4 Practical Recommendations

University administrators should assess the relevance, applicability, complexity, and degree of documentation of research outputs before initiating an industrial transfer project. Where the knowledge is highly technical or tacit, written reports alone may be insufficient. Joint work, demonstrations, training, technical consultation, and continued contact may be required.

Industrial organizations should evaluate whether they possess the personnel, prior knowledge, learning routines, managerial support, and operational resources needed to receive and apply academic knowledge. Identifying these conditions at the beginning of a project can reveal where additional preparation is required.

Universities and industrial partners should also examine the forms of distance between them. Differences in technical language, organizational goals, professional expectations, location, culture, and decision-making procedures can obstruct transfer even when both parties are willing to cooperate. Clear responsibilities, agreed objectives, regular communication, and appropriate transfer channels should therefore be established before implementation.

Policymakers and knowledge-transfer intermediaries may use the framework to organize project assessments and identify recurring barriers across University-Industry partnerships. Because Government Dimension was not statistically significant in the present model, policy decisions should not be based on this result alone. Further evidence is needed to determine whether governmental influence operates indirectly through funding, regulation, intellectual property arrangements, incentives, or intermediary institutions.

The proposed model may be used as an initial diagnostic framework for reviewing the strengths and weaknesses of university-to-industry knowledge transfer projects. Its scores should be interpreted together with project-specific evidence rather than treated as a stand-alone measure of success.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

References

- [1] L. C. S. Silva, C. S. Ten Caten, and S. Gaia, "Conceptual framework of green technology transfer at public university scope Brazilian," *Innov. Green Dev.*, vol. 2, no. 4, p. 100076, 2023. <https://doi.org/10.1016/j.igd.2023.100076>
- [2] J. B. Padilla Bejarano, J. W. Zartha Sossa, C. Ocampo-López, and M. Ramírez-Carmona, "Open innovation: A technology transfer alternative from universities. A systematic literature review," *J. Open Innov. Technol. Mark. Complex.*, vol. 9, no. 3, p. 100090, 2023. <https://doi.org/10.1016/j.joitmc.2023.100090>
- [3] H. Etzkowitz and L. Leydesdorff, "The dynamics of innovation: From National Systems and "Mode 2" to a Triple Helix of university-industry-government relations," *Res. Policy*, vol. 29, no. 2, pp. 109–123, 2000. [https://doi.org/10.1016/S0048-7333\(99\)00055-4](https://doi.org/10.1016/S0048-7333(99)00055-4)
- [4] Y. Zhang, J. Wang, H. Deng, D. Zhang, and Y. Wang, "Developing a multidimensional assessment framework for clean technology transfer potential and its application on the Belt and Road Initiative countries," *J. Clean. Prod.*, vol. 401, p. 136769, 2023. <https://doi.org/10.1016/j.jclepro.2023.136769>
- [5] M. Perkmann and K. Walsh, "University-industry relationships and open innovation: Towards a research agenda," *Int. J. Manag. Rev.*, vol. 9, no. 4, pp. 259–280, 2007. <https://doi.org/10.1111/j.1468-2370.2007.00225.x>
- [6] D. C. Mowery and B. N. Sampat, "Universities in national innovation systems," in *The Oxford Handbook of Innovation*. Oxford, United Kingdom: Oxford University Press, 2005, pp. 209–239.
- [7] J. Bruneel, P. D'Este, and A. Salter, "Investigating the factors that diminish the barriers to university-industry collaboration," *Res. Policy*, vol. 39, no. 7, pp. 858–868, 2010. <https://doi.org/10.1016/j.respol.2010.03.006>
- [8] M. D. Santoro and P. A. Saporito, "Self-interest assumption and relational trust in university-industry knowledge transfers," *IEEE Trans. Eng. Manag.*, vol. 53, no. 3, pp. 335–347, 2006. <https://doi.org/10.1109/TEM.2006.878103>
- [9] P. D'Este and P. Patel, "University-industry linkages in the UK: What are the factors underlying the variety of interactions with industry?" *Res. Policy*, vol. 36, no. 9, pp. 1295–1313, 2007. <https://doi.org/10.1016/j.respol.2007.05.002>
- [10] J. A. Cunningham and A. N. Link, "Fostering university-industry R&D collaborations in European Union countries," *Int. Entrep. Manag. J.*, vol. 11, no. 4, pp. 849–860, 2015. <https://doi.org/10.1007/s11365-014-0317-4>

- [11] M. Perkmann, V. Tartari, M. McKelvey, E. Autio, A. Broström, P. D'Este, R. Fini, A. Geuna, R. Grimaldi, A. Hughes *et al.*, "Academic engagement and commercialisation: A review of the literature on university–industry relations," *Res. Policy*, vol. 42, no. 2, pp. 423–442, 2013. <https://doi.org/10.1016/j.respol.2012.09.007>
- [12] R. Ponds, F. Van Oort, and K. Frenken, "The geographical and institutional proximity of research collaboration," *Pap. Reg. Sci.*, vol. 86, no. 3, pp. 423–443, 2007. <https://doi.org/10.1111/j.1435-5957.2007.00126.x>
- [13] J. L. Cummings and B. S. Teng, "Transferring R&D knowledge: The key factors affecting knowledge transfer success," *J. Eng. Technol. Manage.*, vol. 20, pp. 39–68, 2003. [https://doi.org/10.1016/S0923-4748\(03\)00004-3](https://doi.org/10.1016/S0923-4748(03)00004-3)
- [14] M. Mahdavi Mazdeh, M. A. Shafia, R. Bandarian, and A. Kahrizi, "An ISM approach for analyzing the factors in technology transfer," *Decis. Sci. Lett.*, vol. 4, no. 3, pp. 335–348, 2015.
- [15] T. J. Allen, *Managing the Flow of Technology: Technology Transfer and the Dissemination of Technological Information Within the R&D Organization*. Cambridge, MA, United States: MIT Press, 1977.
- [16] R. Fontana, A. Geuna, and M. Matt, "Factors affecting university–industry R&D projects: The importance of searching, screening and signalling," *Res. Policy*, vol. 35, no. 2, pp. 309–323, 2006. <https://doi.org/10.1016/j.respol.2005.12.001>
- [17] N. Becheikh, S. Ziam, O. Idrissi, Y. Castonguay, and R. Landry, "How to improve knowledge transfer strategies and practices in education? Answers from a systematic literature review," *Res. High. Educ. J.*, vol. 7, pp. 1–21, 2010.
- [18] J. Hemsley-Brown, "Facilitating research utilisation: A cross-sector review of research evidence," *Int. J. Public Sect. Manag.*, vol. 17, no. 6, pp. 534–552, 2004. <https://doi.org/10.1108/09513550410554805>
- [19] Y. F. Barnard, G. J. Veldhuis, and J. C. G. M. van Rooij, "Evaluation in practice: Identifying factors for improving transfer of training in technical domains," *Stud. Educ. Eval.*, vol. 27, no. 3, pp. 269–290, 2001.
- [20] D. Schartinger, A. Schibany, and H. Gassler, "Interactive relations between universities and firms: Empirical evidence for Austria," *J. Technol. Transf.*, vol. 26, no. 3, pp. 255–268, 2001. <https://doi.org/10.1023/A:1011110207885>
- [21] J. Bercovitz and M. Feldman, "Entrepreneurial universities and technology transfer: A conceptual framework for understanding knowledge-based economic development," *J. Technol. Transf.*, vol. 31, no. 1, pp. 175–188, 2006. <https://doi.org/10.1007/s10961-005-5029-z>
- [22] Y. Duan, W. Nie, and E. Coakes, "Identifying key factors affecting transnational knowledge transfer," *Inf. Manag.*, vol. 47, no. 7–8, pp. 356–363, 2010. <https://doi.org/10.1016/j.im.2010.08.003>
- [23] W. M. Cohen and D. A. Levinthal, "Absorptive capacity: A new perspective on learning and innovation," *Admin. Sci. Quart.*, vol. 35, no. 1, pp. 128–152, 1990. <https://doi.org/10.2307/2393553>